

vulmshelp.com

Paper No 19

FINALTERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Ref No:

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

Scatterplots are used only for

- ▶ quantitative variables
- ▶ **multivariables**
- ▶ surplus variable
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

Changed with the DEMO VERSION of CAD-KAS PDF-Editor (<http://www.cadkas.com>).
If the CORRELATION function returns the #DIV/0! error value, what is the possible reason of the error?

- ▶ Array1 and Array2 have different number of data points.
- ▶ **Either Array1 or array2 is empty.**
- ▶ Array or reference argument contains text, logical values or empty cells.
- ▶ The arguments are names, arrays, or references that contain numbers.

Question No: 3 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

- ▶ **1260**
- ▶ 210
- ▶ $10C6 * 6!$
- ▶ $10C5 * 6$

<http://www.onestopmba.com/cat-tips/materials/maths/algebra/permutation/answer-one.asp>

Question No: 4 (Marks: 1) - Please choose one

An insurance company wants to predict sales from the amount of money they spend on advertising. Which would be the independent variable?

- ▶ (i) sales
- ▶ **(ii) advertising**

- ▶ (iii) insufficient information to decide
- ▶ (iv) Both (i) and (ii) are correct.

Question No: 5 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- ▶ **bar graph**
- ▶ pie graph
- ▶ pictograph
- ▶ line graph

Question No: 6 (Marks: 1) - Please choose one

In a positively skewed distribution

- ▶ The mean, median, and mode are all equal.
- ▶ **The mean is larger than the median**
- ▶ The median is larger than the mean.
- ▶ The standard deviation must be larger than the mean or the median.

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is not a component of the multiplicative time series model?

- ▶ trend
- ▶ irregular variation
- ▶ **regression trend**
- ▶ seasonality
- ▶ cyclicity

vulmshelp.com

Question No: 8 (Marks: 1) - Please choose one

How many number of times will the digit '7' be written when listing the integers from 1 to 1000?

- ▶ 271
- ▶ **300**
- ▶ 252
- ▶ 304

Question No: 9 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

- ▶ 6723
- ▶ 6725
- ▶ **6720**

► 6721

Question No: 10 (Marks: 1) - Please choose one

The moving averages represent -----

► **Time series variations**

- Co-efficient of variations
- Statistical Dispersion
- Absolute deviation

Question No: 11 (Marks: 1) - Please choose one

The class frequency is

► **The number of observations in each class**

- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

Question No: 12 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the standard deviation is approximately

Changed with the DEMO VERSION of CAD-KAS PDF-Editor (<http://www.cadkas.com>).

► **4.75**

- 4.55
- 3.03
- 3.75

Question No: 13 (Marks: 1) - Please choose one

The experimental region is the range of the previously observed values of the dependent variable.

► **False**

► True

Question No: 14 (Marks: 1) - Please choose one

Which of the following is not a violation of the independence assumption?

- Negative autocorrelation
- A pattern of cyclical error terms over time
- Positive autocorrelation
- A pattern of alternating error terms overtime
- **A random pattern of error terms over time**

Question No: 15 (Marks: 1) - Please choose one

vulmshelp.com

All of the following are assumptions of the error terms in the simple linear regression model except

- ▶ normality.
- ▶ error terms with a mean of zero.
- ▶ constant variance.
- ▶ **variance of one.**

Question No: 16 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- ▶ slope, intercept
- ▶ **intercept, slope**
- ▶ **slope, regression coefficient**
- ▶ radius, intercept

Question No: 17 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above

Question No: 18 (Marks: 1) - Please choose one

If A and B are two mutually exclusive events, then

- ▶ $P(A \cup B) = P(A) \cdot P(B)$
- ▶ $P(A \cap B) = P(A) + P(B)$
- ▶ **$P(A \cup B) = P(A) + P(B)$**
- ▶ $P(A \cup B \cup C) = P(A) \cdot P(B)$

Question No: 19 (Marks: 1) - Please choose one

The equation for the correlation coefficient is _____ where

- ▶ _____
- ▶ _____
- ▶ _____
- ▶ _____

Question No: 20 (Marks: 1) - Please choose one

After the merchant buys merchandise, it is sold at a higher price called the _____

- ▶ Sale price
- ▶ Revenue discount
- ▶ **Selling price**
- ▶ Cost price

Question No: 21 (Marks: 1) - Please choose one

A reduction of the amount due on an invoice is called a _____ .

- ▶ Trade discount
- ▶ **Net discount**
- ▶ Cash discount
- ▶ Unearned discount

vulmshelp.com

Question No: 22 (Marks: 1) - Please choose one

If the basic salary of an employee is 13000 what is the amount of allowances he is getting for his conveyance?

▶ **325**

Changed with the DEMO VERSION of CAD-KAS PDF-Editor (<http://www.cadkas.com>).

- ▶ 260
- ▶ 765
- ▶ 500

Question No: 23 (Marks: 1) - Please choose one

The minimum value of the correlation coefficient r can be

- ▶ -infinity
- ▶ **0**
- ▶ -1
- ▶ 1

Question No: 24 (Marks: 1) - Please choose one

Total Provident Fund added to the employee's fund is ----- of the basic salary.

- ▶ 1 / 11 th
- ▶ **2 / 11 th**
- ▶ 9.09 %
- ▶ 9.99%

Question No: 25 (Marks: 1) - Please choose one

If I is an identity matrix then it must also be a

- ▶ rectangular matrix
- ▶ row matrix

- ▶ column matrix
- ▶ **scalar matrix**

Question No: 26 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ▶ **=AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)**
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

If A is the matrix of dimension $n \times m$ and I is the identity matrix of dimension $n \times n$. Then which of the following is true

- ▶ $AI = I$
- ▶ **$AI = A$**
- ▶ $A + I = A$

Question No: 28 (Marks: 1) - Please choose one

If the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

- ▶ 33.33%
- ▶ **25%**
- ▶ 20%
- ▶ 10%

Question No: 29 (Marks: 1) - Please choose one

The net price of a computer table is 2500 and list price 3000. The trade discount will be

- _ 13%
- _ 17%**
- _ 21%
- _ 25%--

Question No: 30 (Marks: 1) - Please choose one

Ogives at different values determine

- ☐ limits .
- ☐ frequencies.
- ☐ cumulative frequencies.
- ☒ both (2) & (3).

Question No: 31 (Marks: 1) - Please choose one

While using Frequency function ,one always selects
one cell more than data array.

- ☒ one cell more than bins array .
- ☐ at most 20 cells.
- ☐ random number of cells

Question No: 32 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

- ☐ m+n
- ☐ m-n
- ☒ m*n
- ☐ n/m

Question No: 33 (Marks: 1) - Please choose one

The variance of the 3,3,3,3,3 is

- ☒ Zero.
- ☐ One.
- ☐ Twice the standard standard deviation.
- ☐ Half of the range

Changed with the DEMO VERSION of CAD-KAS PDF-Editor (<http://www.cadkas.com>).

Question No: 34 (Marks: 1) - Please choose one

Which of the following is most probably is an example of impossible event while tossing of a coin

- ☐ Coming of head
- ☐ Coming of tail
- ☒ Coming of erected coin
- ☐ None of these

Question No: 35 (Marks: 1) - Please choose one

For two tail test, when

$\alpha = 0.10$

the value of Z is

- ☐ ± 1.96
- ☒ ± 1.645
- ☐ ± 2.326
- ☐ ± 2.575

Question No: 36 (Marks: 1) - Please choose one

For the set of data 1, 2, 3, 4, 5, 2, 1, 6, 8, the mode is given by

- ☐ 1
- ☐ 2
- ☒ 1 and 2
- ☐ 3

Question No: 37 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

- ☒ True
- ☐ False

Question No: 38 (Marks: 1) - Please choose one

Linear trend is calculated as $Y_t = 30.5 + 0.55t$. The trend projection for period 15 is

- ☐ 11.25
- ☐ 28.50
- ☒ 38.75
- ☐ 44.25

Question No: 39 (Marks: 1) - Please choose one

.....is a deduction from the list price of goods provided by a business in return for payment within a specified time.

- ☒ trade discount
- ☐ cash discount
- ☐ credit discount
- ☐ none of these

Question No: 40 (Marks: 1) - Please choose one

Ogive of a statistical data can be drawn by

- ☒ a) using the cumulative frequency of the distribution
- ☐ b) frequency of the distribution
- ☐ c) both (a) & (b)
- ☐ d) None of these.

Question No: 41 (Marks: 2)

Find harmonic mean (HM) of 10,12,14,17

Question No: 42 (Marks: 2)

If you toss a die and observe the number of dots that appear on top face then write the events that the even number occurs.

Solution:

Number of Possible outcomes = 6

Number of Events = 3

$P = \text{Number of events} / \text{Number of all Possible outcomes.}$

$P = 3/6$

Question No: 43 (Marks: 2)

Define Cumulative Poisson distribution.

Solution:

A cumulative Poisson distribution is used to calculate the probability of getting atleast n successes in a Poisson experiment. Here, n is the Poisson random variable which refers to the number of success.

Formula:

$$P(x < n) = P(x = 0) + P(x = 1) + \dots + P(x = n)$$

where, $P(x = 0)$ and $P(x = 1)$ is calculated using Poisson distribution formula.

Question No: 44 (Marks: 3)

Four friends take an IQ test. Their scores are 96, 100, 106, 114. Which of the following statements is true? Give reason.

- I. The mean is 103.
- II. The mean is 104.
- III. The median is 100.
- IV. The median is 106.

- (A) I only
- (B) II only
- (C) III only
- (D) IV only
- (E) None is true

Solution:

The correct answer is (B). The mean score is computed from the equation:

$$\text{Mean score} = \frac{\sum x}{n} = \frac{(96 + 100 + 106 + 114)}{4} = 104$$

Since there are an even number of scores (4 scores), the median is the average of the two middle scores. Thus, the median is $(100 + 106) / 2 = 103$

Question No: 45 (Marks: 3)

What is the probability that a bag filled by the machine weighs less than 515 g?

$$z = (515 - 510) / 2.5 = 2.0$$

(Use the table given below)

vulmshelp.com